



Ahmed Aziz Ben Haj Hmida

Data Science Graduate (EPFL) with a Minor in Management, Technology & Entrepreneurship | Specialized in leveraging AI and Analytics to drive industrial R&D and Business Innovation.

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Education

EPFL – École Polytechnique Fédérale de Lausanne

M.Sc. in Data Science

Minor in Management, Technology & Entrepreneurship (MTE)

GPA: 5.3/6

Relevant Coursework: Applied Data Analysis, Data Science for Business, Finance, Microeconomics, Project Management & Risk Analysis, Value Chain Management, Economic Growth & Sustainability

Lausanne, Switzerland

Sep 2023 – Mar 2026

Technical University of Denmark (DTU)

3rd Year of Bachelor – Exchange

GPA: 5.35/6

Lyngby, Denmark

Aug 2022 – Jul 2023

EPFL – École Polytechnique Fédérale de Lausanne

B.Sc. in Communication Systems

GPA: 5.1/6

Lausanne, Switzerland

Sep 2020 – Jul 2023

Pioneer School of Nabeul

Baccalaureate in Mathematics

Nabeul, Tunisia

Sep 2018 – Jun 2019

Professional Experience

Rollomatic SA

Master's Thesis: Strategic Retrieval System for CNC Tools (Grade 6/6)

Le Landeron, Switzerland

Aug 2025 – Mar 2026

- Engineered an end-to-end similarity search system that reduced search latency by **96%**, drastically accelerating tool design cycles.
- Improved retrieval accuracy by **11%**, enabling design reuse and reducing manufacturing R&D costs.
- Integrated the system into industrial workflows, translating complex technical specifications into actionable tools.

Supervisors: Prof. Jean-Philippe Thiran - jean-philippe.thiran@epfl.ch, Emna Maghrebi - e.maghrebi@rollomatic.ch

Rollomatic SA

Internship: Data-Driven Optimization for Laser Machining

Le Landeron, Switzerland

Feb 2025 – Aug 2025

- Led the initial research workstream to identify inefficiencies in tool retrieval processes for laser machining units.
- Developed a prototype hybrid search pipeline integrating deep visual embeddings to streamline industrial workflows.
- Collaborated with R&D stakeholders to define KPIs, setting the strategic foundation for the subsequent production-level system.

Supervisor: Emna Maghrebi - e.maghrebi@rollomatic.ch

Research Projects

Discovering Promising New Startups Through Evaluating Teams' Potentials

Sep 2023 – Dec 2023

EPFL ENTC Lab

- Harnessed ML methods to derive speakers' MBTI personalities and emotions during entrepreneurial discussions.
- Built predictive approaches to identify promising teams with high startup success potential.

Supervisor: Prof. Davide Bavato

Unsupervised Reverse Knowledge Distillation and Model Merging

Aug 2024 – Feb 2025

EPFL LTS₄ Lab

- Transferred knowledge from compact task-specific models into large-scale architectures.
- Applied model merging techniques to improve niche-task performance while preserving broad generative capabilities.

Supervisor: Prof. Pascal Frossard - pascal.frossard@epfl.ch, Ortal Senouf

Technical Skills

Programming Languages: Python, Java, SQL, MATLAB, Scala

Business & Strategy: Excel, PowerPoint, Financial Modeling, Project Management, Risk Analysis, Power BI

Data Analytics & ML: PyTorch, Scikit-learn, RAG Pipelines, Agentic AI, PostgreSQL

Tools & DevOps: Git, Docker, GitHub Actions, Kubernetes, FastAPI, W&B, GCP, AWS, Streamlit

Academic Projects

LLM as a Teaching Assistant: Leveraging LLM for MCQ Answering Feb 2024 – Jun 2024

- Fine-tuned Qwen1.5 using SFT and Direct Preference Optimization with RAG and 4-bit quantization.
- Developed a chatbot interface and evaluation suite using LLM-as-a-judge frameworks and cross-validation.

Machine Learning Operations Jan 2023 – Mar 2023

- Deployed scalable ML pipelines using PyTorch, Docker, Kubernetes, CI/CD, GCP, AWS, MLflow, and DVC.

Exploring the Transition from Novice to Expert in RL Policies for Motor Skill Feb 2024 – Jun 2024

EPFL Alexander Mathis Lab

- Trained expert reinforcement learning agents on Myosuite baoding ball tasks.
- Distilled motor strategies into novice agents to improve learning efficiency and performance.

Coin Detector and Counter Feb 2024 – Jun 2024

- Used segmentation techniques and deep learning models (CNNs, Vision Transformers, ResNet50) for accurate coin recognition.

Certificates

Structuring Machine Learning Projects – DeepLearning.AI Sep 2023 - Oct 2023

Certificate in Artificial Intelligence – GOMYCODE Jun 2023 – Sep 2023

Associations

EPFL Coaching – Coach of CMS section Sep 2023 – Jun 2024

Rotary International – Member Jul 2018 – Apr 2019

Tunisian Red Crescent – Volunteer May 2018 – Apr 2019

Languages

Arabic/French: Native | English: C1 | German: A2